

Free products and AQFT

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Abstract

We review the free product construction of von Neumann algebras, its application to a question in Algebraic Quantum Field Theory (AQFT) and an application of AQFT techniques to a question of free products. We show the existence of half-sided modular inclusions with trivial relative commutant and nontriviality of relative commutant for an inclusion of free product von Neumann algebras.

1 Introduction

Given a pair of von Neumann algebras $\mathcal{M}_1, \mathcal{M}_2$ and a pair of normal states ω_1, ω_2 , one can construct the free product von Neumann algebra $\mathcal{M} = (\mathcal{M}_1, \omega_1) \star (\mathcal{M}_2, \omega_2)$ that contains both $\mathcal{M}_1, \mathcal{M}_2$ in such a way that they do not commute in the highest degree, up to some “relations” provided by the states ω_1, ω_2 (see [VDN92]). It is also possible to consider the free product of an indexed family $\{(\mathcal{M}_\kappa, \omega_\kappa)\}_{\kappa \in K}$. Fundamental properties of the free product $\mathcal{M}$, such as factoriality and type classification have been established recently, see [Ued11] and references therein.

If we have a family of inclusions of von Neumann algebras $\{(\mathcal{N}_\kappa \subset \mathcal{M}_\kappa, \omega_\kappa)\}_{\kappa \in K}$, we obtain an inclusion of free product von Neumann algebras $\mathcal{N} \subset \mathcal{M}$. It is a natural question whether the relative commutant $\mathcal{M} \cap \mathcal{N}'$ is trivial or how large it is, but there have been few results for the cases where there is no conditional expectation from $\mathcal{M}_\kappa$ to $\mathcal{N}_\kappa$. We showed in [LTU17] that the relative commutant $\mathcal{M} \cap \mathcal{N}'$ changes dramatically depending on whether K is finite or infinite, and whether the family $\{(\mathcal{N}_\kappa \subset \mathcal{M}_\kappa, \omega_\kappa)\}_{\kappa \in K}$ is identical or not.

The question of relative commutant is important in the operator-algebraic approach to quantum field theory, or Algebraic QFT (AQFT). In AQFT, we consider a family of von Neumann algebras $\{\mathcal{A}(O)\}$ associated with spacetime regions $\{O\}$ satisfying isotony, namely if $O_1 \subset O_2$ then $\mathcal{A}(O_1) \subset \mathcal{A}(O_2)$. The relative commutant $\mathcal{A}(O_2) \cap \mathcal{A}(O_1)'$ contains $\mathcal{A}(O)$ if O is contained in O_2 and is spacelike to O_1 . A way to construct an example in one-dimensional spacetime (Möbius covariant net) is to prepare a pair of von Neumann algebras $\mathcal{N} \subset \mathcal{M}$ satisfying certain conditions (including half-sided modular inclusion, see below) and prove

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that $\mathcal{M} \cap \mathcal{N}'$ is sufficiently large [GLW98]. There are many such pairs $\mathcal{N} \subset \mathcal{M}$, but it was difficult to determine the relative commutant. By free product, we provide a first example of half-sided modular inclusion with trivial relative commutant $\mathcal{M} \cap \mathcal{N}' = \mathbb{C}\mathbb{1}$ [LTU17].

On the other hand, there are sufficient conditions that assure that the relative commutant $\mathcal{M} \cap \mathcal{N}'$ is large. One of them is the L^2 -nuclearity condition, widely known in the context of AQFT. An example from AQFT provides a free product inclusion $\mathcal{N} \subset \mathcal{M}$ with nontrivial relative commutant $\mathcal{M} \cap \mathcal{N}'$ [LTU17]. This is a very rare result: when the relative commutant can be determined, it is often trivial. AQFT provides such a rare technique.

We recall the fundamental notions of free product and AQFT and give an overview of these results.

2 Free product of von Neumann algebras

Let K be an index set and $\{\mathcal{H}_\kappa\}_{\kappa \in K}$ a family of Hilbert spaces, $\mathcal{M}_\kappa$ von Neumann algebras acting on $\mathcal{H}_\kappa$ and $\Omega_\kappa \in \mathcal{H}_\kappa$ a cyclic and separating vector for $\mathcal{M}_\kappa$. The free product of $\{(\mathcal{M}_\kappa, \omega_\kappa)\}_{\kappa \in K}$, where $\omega_\kappa(\cdot) = \langle \Omega_\kappa, \cdot \Omega_\kappa \rangle$ is defined on the Hilbert space

$$\mathcal{H} = \mathbb{C}\Omega \oplus \bigoplus_{n \geq 1} \bigoplus_{\substack{\kappa_j \neq \kappa_{j+1} \\ 1 \leq j \leq n-1}} \mathcal{H}_{\kappa_1}^\circ \bar{\otimes} \cdots \bar{\otimes} \mathcal{H}_{\kappa_n}^\circ \quad \text{with} \quad \mathcal{H}_\kappa^\circ := \mathcal{H}_\kappa \ominus \mathbb{C}\Omega_\kappa,$$

where Ω spans a one-dimensional subspace $\mathbb{C}\Omega$. The tensor factors $\mathcal{H}_\kappa^\circ$ appear in all possible alternating way. For each index κ , this can be rearranged as follows

$$\begin{aligned} \mathcal{H} &= \bigoplus_{n \geq 1} \bigoplus_{\substack{\kappa \neq \kappa_2 \\ \kappa_j \neq \kappa_{j+1} \\ 2 \leq j \leq n-1}} (\mathcal{H}_\kappa^\circ \bar{\otimes} \mathcal{H}_{\kappa_2}^\circ \bar{\otimes} \cdots \bar{\otimes} \mathcal{H}_{\kappa_n}^\circ) \oplus (\mathcal{H}_{\kappa_2}^\circ \bar{\otimes} \cdots \bar{\otimes} \mathcal{H}_{\kappa_n}^\circ) \\ &\cong \mathcal{H}_\kappa \otimes \mathcal{H}'_\kappa, \end{aligned}$$

where the case $n = 1$ is $\mathcal{H}_\kappa^\circ \oplus \mathbb{C}\Omega \cong \mathcal{H}_\kappa$ and $\mathcal{H}'_\kappa$ is a direct sum of all $\mathcal{H}_{\kappa_2}^\circ \bar{\otimes} \cdots \bar{\otimes} \mathcal{H}_{\kappa_n}^\circ$ where $\kappa \neq \kappa_2$. By this identification, $\mathcal{B}(\mathcal{H}_\kappa)$ acts from the left on the component $\mathcal{H}_\kappa$. This is called the left action and we denote it by λ_κ . Similarly, we denote the right action by ρ_κ . The von Neumann algebra

$$\mathcal{M} = \bigvee_{\kappa \in K} \lambda_\kappa(\mathcal{M}_\kappa)$$

is called the **free product von Neumann algebra** of the family $\{(\mathcal{M}_\kappa, \omega_\kappa)\}_{\kappa \in K}$. The vector Ω is cyclic and separating for $\mathcal{M}$. It is also possible to characterize the free product von Neumann algebra abstractly, see [Voi85, §1.5].

One of the characteristic properties is freely independence: let us denote $\omega(\cdot) = \langle \Omega, \cdot \Omega \rangle$. The von Neumann algebra $\mathcal{M}$ contains isomorphic copies of $\mathcal{M}_\kappa$'s and it holds that

$$\omega(\lambda_{\kappa_1}(x_1) \cdots \lambda_{\kappa_n}(x_n)) = 0$$

for $x_j \in \text{Ker}(\omega_{\kappa_j}) \cap \mathcal{M}_{\kappa_j}$ with $\kappa_j \neq \kappa_{j+1}$ ($1 \leq j \leq n-1$) and $n \geq 1$.

We introduce the unitary involution $Z : \mathcal{H} \rightarrow \mathcal{H}$ by $Z\Omega = \Omega$ and

$$Z(\xi_1 \otimes \cdots \otimes \xi_n) = \xi_n \otimes \cdots \otimes \xi_1 \quad \text{for} \quad \xi_1 \otimes \cdots \otimes \xi_n \in \mathcal{H}_{\kappa_1}^\circ \bar{\otimes} \cdots \bar{\otimes} \mathcal{H}_{\kappa_n}^\circ \subset \mathcal{H},$$

namely, Z reverses the order. As remarked in [Voi85, §1.7], Z exchanges the left and right actions $\lambda_\kappa, \rho_\kappa$, and we have $\mathcal{M}' = \bigvee_{\kappa \in K} \rho'_\kappa(\mathcal{M}_\kappa)$ [Voi85, Proposition 1.9].

The modular objects for $(\mathcal{M}, \Omega)$ can be determined from those for $(\mathcal{M}_\kappa, \Omega_\kappa)$:

$$\sigma_{\mathcal{M}}^t \circ \lambda_\kappa = \lambda_\kappa \circ \sigma_{\mathcal{M}_\kappa}^t$$

for every $\kappa \in K$, where $\sigma_{\mathcal{M}_\kappa}^t$ is the modular automorphism of $\mathcal{M}_\kappa$ associated with ω_κ [Bar95, Lemma 1][Dyk94, Theorem 1]. Furthermore, for a family $\{T_\kappa\}_{\kappa \in K}$ of bounded operators on $\mathcal{H}_\kappa$ with $\|T_\kappa\| \leq 1$ that preserve $\mathcal{H}_\kappa^\circ$, we introduce

$$\star_{\kappa \in K} T_\kappa := T_\Omega \oplus \bigoplus_{n \geq 1} \bigoplus_{\substack{\kappa_j \neq \kappa_{j+1} \\ 1 \leq j \leq n-1}} T_{\kappa_1}^\circ \otimes \cdots \otimes T_{\kappa_n}^\circ \in \mathcal{B}(\mathcal{H}) \quad \text{with} \quad T_\kappa^\circ := T_\kappa \upharpoonright_{\mathcal{H}_\kappa^\circ},$$

where $T_\Omega = \mathbb{1}_\mathbb{C}$ if T_κ 's are linear and $T_\Omega = J_\mathbb{C}$ (the complex conjugation) if T_κ 's are antilinear. With this notation, we have

$$\Delta_{\mathcal{M}}^{it} = \star_{\kappa \in K} \Delta_{\mathcal{M}_\kappa}^{it}, \quad J_{\mathcal{M}} = Z(\star_{\kappa \in K} J_{\mathcal{M}_\kappa}) = (\star_{\kappa \in K} J_{\mathcal{M}_\kappa})Z,$$

where $\Delta_{\mathcal{M}_\kappa}$ and $J_{\mathcal{M}_\kappa}$ are the modular operator and the modular conjugation for $\mathcal{M}_\kappa$ with respect to Ω_κ .

3 Half-sided modular inclusions and algebraic QFT

Half-sided modular inclusion In algebraic QFT, important roles are played by certain inclusions of von Neumann algebras that are not very much investigated in the pure mathematics literature. Half-sided modular inclusions are such an example. A **half-sided modular inclusion** (HSMI) $(\mathcal{N} \subset \mathcal{M}, \Omega)$ is an inclusion $\mathcal{N} \subset \mathcal{M}$ of von Neumann algebras and a vector Ω which is cyclic and separating for $\mathcal{N}$ and $\mathcal{M}$ such that $\sigma_t^{\mathcal{M}}(\mathcal{N}) \subset \mathcal{N}$ for $t \geq 0$, where $\sigma_t^{\mathcal{M}}$ is the modular automorphism group of $\mathcal{M}$ with respect to Ω . One obtains a HSMI by considering a certain inclusion of local algebras in AQFT, as we review below.

A HSMI $(\mathcal{N} \subset \mathcal{M}, \Omega)$ is said to be **standard** if Ω is cyclic and separating for $\mathcal{N}' \cap \mathcal{M}$. This case is of great interest, because from a standard HSMI one can construct a Möbius covariant net, see below. This is why we need to study relative commutant.

Möbius covariant net A Möbius covariant Haag-Kastler net, or **Möbius covariant net** on S^1 for short, is a mathematical realization of a chiral component of two-dimensional conformal field theory. The Möbius group $\text{Möb} := \text{SL}(2, \mathbb{R})/\mathbb{Z}_2 \cong \text{PSU}(1, 1)$ acts on S^1 by linear fractional transformations. A Möbius covariant net is a triple $(\mathcal{A}, U, \Omega)$ on a Hilbert space $\mathcal{H}$, where $\{\mathcal{A}(I)\}$ are a family of von Neumann algebras on a fixed Hilbert space $\mathcal{H}$ parametrized by open proper intervals $\{I\}$ of S^1 , a strongly continuous representation U on $\mathcal{H}$ of Möb and a vector $\Omega \in \mathcal{H}$ satisfying the (not necessarily local) **Haag-Kastler axioms**: isotony, Möbius covariance, positivity of the energy and uniqueness of vacuum. A Möbius covariant net $\mathcal{A}$ is said to be **local** if $\mathcal{A}(I_1)$ and $\mathcal{A}(I_2)$ commute for disjoint I_1, I_2 . Usually locality is included in the Haag-Kastler axioms, but we treat it separately, as nonlocal Möbius covariant nets appear naturally in our construction, see [LTU17] for the details.

The interplay between AQFT and the modular theory shows up immediately: $\Delta_I^{it} := \Delta_{\mathcal{A}(I)}^{it} = U(\Lambda_I(-2\pi t))$ holds (the Bisognano-Wichmann property), where $\Delta_{\mathcal{A}(I)}$ is the modular operator of $\mathcal{A}(I)$ with respect to Ω and Λ_I is the one-parameter group in Möb that preserves I , see [DLR01, Proposition 2.2(ii)]. It also follows that the algebras $\mathcal{A}(I)$ are of type III₁ (unless the Hilbert space is one-dimensional) [DLR01, Proposition 2.4(ii)].

If $(\mathcal{N} \subset \mathcal{M}, \Omega)$ is a standard HSMI, then one can construct a local Möbius covariant net (possibly without uniqueness of vacuum) as follows. We identify S^1 and the one-point compactification of $\mathbb{R}$ by stereographic projection. The Möbius group Möb is generated by three one-parameter groups that preserve the intervals $(1, \infty)$, $(0, \infty)$ and $(0, 1)$, respectively, under the above identification. Then we define $\mathcal{A}((1, \infty)) = \mathcal{N}$, $\mathcal{A}((0, \infty)) = \mathcal{M}$, and U is constructed from the modular groups $\Delta_{\mathcal{N}}^{it}, \Delta_{\mathcal{M}}^{it}, \Delta_{\mathcal{M} \cap \mathcal{N}'}^{it}$ [GLW98, Corollary 1.7].

Techniques for relative commutant It is not easy to determine the size of the relative commutant $\mathcal{M} \cap \mathcal{N}'$, but in some situation it can be proven to be large. One of such cases is the **split** inclusion, namely, an inclusion $\mathcal{N} \subset \mathcal{M}$ of von Neumann algebras is split if there is a type I factor $\mathcal{R}$ such that $\mathcal{N} \subset \mathcal{R} \subset \mathcal{M}$. If $\mathcal{N}$ and $\mathcal{M}$ are furthermore type III algebras on a separable Hilbert space $\mathcal{H}$, then the split property assures that the relative commutant $\mathcal{N}' \cap \mathcal{M}$ is again of type III, in particular it is nontrivial.

The split property is still an abstract property to check in examples, but there are several handier sufficient conditions for the split property. Let Ω be cyclic and separating for both of $\mathcal{M}$ and $\mathcal{N}$, and $\Delta_{\mathcal{N}}, \Delta_{\mathcal{M}}$ be the corresponding modular operators. If $\Delta_{\mathcal{M}}^{\frac{1}{4}} \Delta_{\mathcal{N}}^{-\frac{1}{4}}$ is nuclear (the **L^2 -nuclearity** condition), then the split property follows [BDL07].

Let us turn to a Möbius covariant net $(\mathcal{A}, U, \Omega)$, not necessarily local. In the Möbius group Möb there is the one-parameter group of rotations. Let L_0 be the generator of rotations in the representation U . If $\text{Tr } e^{-sL_0} < \infty$ for some s (**trace class property**), then $\mathcal{A}(I) \subset \mathcal{A}(\tilde{I})$ is split for I sufficiently small with respect to $\tilde{I}$ in terms of “conformal distance” [BDL07, Corollary 6.4]. In most examples of conformal net, the dimensions of the eigenspaces of L_0 can be estimated, hence trace class property can be easily checked.

4 An application of free product to AQFT and vice versa

As we know that the modular objects of the free product von Neumann algebras can be computed from those of the components, it is immediate that, if $\{(\mathcal{N}_\kappa \subset \mathcal{M}_\kappa, \omega_\kappa)\}_{\kappa \in K}$ is a family of HSMIs, then so is the free product $(\mathcal{N} \subset \mathcal{M}, \Omega)$, where $\mathcal{N} = \star_{\kappa \in K} (\mathcal{N}_\kappa, \omega_\kappa) \subset \mathcal{M} = \star_{\kappa \in K} (\mathcal{M}_\kappa, \omega_\kappa)$.

The first of our main results is a **HSMI with trivial relative commutant**.

Theorem 4.1. *Let $\{(\mathcal{N}_\kappa \subset \mathcal{M}_\kappa, \omega_\kappa)\}_{\kappa \in K}$ be a infinite family of the copies of a single HSMI $(\mathcal{N}_0 \subset \mathcal{M}_0, \Omega)$. Then for the free product HSMI $(\mathcal{N} \subset \mathcal{M}, \Omega)$ it holds that $\mathcal{M} \cap \mathcal{N}' = \mathbb{C}\mathbb{1}$.*

We usually assume that the canonical endomorphism acts ergodically on the original HSMI $(\mathcal{N}_0 \subset \mathcal{M}_0, \Omega)$, and in this case the same holds for $(\mathcal{N} \subset \mathcal{M}, \Omega)$. For the relative commutant, we do not use the assumption that $\{(\mathcal{N}_\kappa \subset \mathcal{M}_\kappa, \omega_\kappa)\}_{\kappa \in K}$ are HSMIs, but it

is enough that $\mathcal{N}_k$ are nontrivial. The both assumptions that K is infinite and $\{(\mathcal{N}_\kappa \subset \mathcal{M}_\kappa, \omega_\kappa)\}_{\kappa \in K}$ are identical are essential in the proof. We exploit the conditional expectation $E_{K_1} : \mathcal{M} \rightarrow \mathcal{M}_{K_1} = \star_{\kappa \in K_1} (\mathcal{M}_\kappa, \omega_\kappa)$ for a subset K_1 of K and the fact that $E_{K_1}(x) \rightarrow x$ as $K_1 \nearrow K$, where it is crucial that K is an infinite set.

The second and the third of our results are an **inclusion of free product von Neumann algebras with nontrivial relative commutant**. Let $\{(\mathcal{A}_\kappa, U_\kappa, \Omega_\kappa)\}_{\kappa \in K}$ be a family of local Möbius covariant nets. Then one can make the free product Möbius covariant net $(\mathcal{A}, U, \Omega)$, where $\mathcal{A}(I) = \star_{\kappa \in K} (\mathcal{A}_\kappa(I), \omega_\kappa)$ and U defined by second quantization, which fails to satisfy locality. Yet, we have the following.

Theorem 4.2. *Let $\{(\mathcal{A}_\kappa, U_\kappa, \Omega_\kappa)\}_{\kappa \in K}$ be a finite family of local Möbius covariant nets with trace class property for arbitrarily small s . Then the free product Möbius covariant net $(\mathcal{A}, U, \Omega)$ satisfies trace class property for sufficiently large s , hence the inclusion of free product von Neumann algebras $\mathcal{A}(I) \subset \mathcal{A}(\tilde{I})$ has a type III relative commutant for some pair $I \subset \tilde{I}$.*

The proof is done by estimating the trace of e^{-sL_0} . This can be done because L_0 has the second quantized form, and all its eigenvectors are alternating tensor product of the eigenvectors of $L_{0,\kappa}$. One cannot apply this technique to HSMI, because the HSMI $\mathcal{A}((1, \infty)) \subset \mathcal{A}((0, \infty))$ coming from a Möbius covariant net is never split (see the argument of [Buc74, P.292(b)]).

Next, let $(\mathcal{A}_0, \mathcal{U}_0, \Omega_0)$ be a Möbius covariant net with L^2 -nuclearity property. The nuclearity norm for the pair $I \subset \tilde{I}$ decreases to 0 as the conformal distance increases. Let $I_\kappa, \kappa \in K$ be a sequence of intervals that shrinks sufficiently fast. Then we can show that $\star_{\kappa \in K} (\mathcal{A}_0(I_\kappa), \Omega_0) \subset \star_{\kappa \in K} (\mathcal{A}_0(\tilde{I}), \Omega_0)$ has nontrivial relative commutant, this time by applying L^2 -nuclearity condition.

These examples show that both of the assumptions of Theorem 4.1 (that K is infinite and all $(\mathcal{N}_k \subset \mathcal{M}_k, \Omega_k)$ are identical) are essential.

5 Open problems

In Theorem 4.1 we constructed a free product HSMI with trivial relative commutant, while we also constructed inclusions (not HSMI) of free product von Neumann algebras with large relative commutant. A natural question is the following.

Problem 5.1. *Let $\{(\mathcal{N}_\kappa \subset \mathcal{M}_\kappa, \Omega_\kappa)\}_{\kappa \in K}$ be a finite family of HSMIs. Can the free product HSMI $(\mathcal{N} \subset \mathcal{M}, \Omega)$ have nontrivial relative commutant?*

An affirmative answer would provide a new Möbius covariant net through [GLW98, Corollary 1.7][BLM11, Lemma 5.1, Theorem 5.2], hence will be of great interest. Yet, such a HSMI cannot be split [Buc74, P.292(b)], hence new techniques are required.

Similarly, one can construct a two-dimensional Haag-Kastler net from a von Neumann algebra, a representation T of $\mathbb{R}^2$ and a vector Ω satisfying certain conditions (Borchers triple). One can make the free product of Borchers triples, and we would like to the following.

Problem 5.2. *Let $\{(\mathcal{M}_\kappa, T_\kappa, \Omega_\kappa)\}_{\kappa \in K}$ be a finite family of Borchers triples. Can one construct a nontrivial Haag-Kastler net from the free product Borchers triple $(\mathcal{M}, T, \Omega)$?*

Here one might use the split property. A sufficient condition is modular nuclearity condition [BDL90, BL04]: for $\mathcal{N} \subset \mathcal{M}$, Ω a cyclic and separating vector for $\mathcal{N}, \mathcal{M}$, the map $\mathcal{N} \ni x \mapsto \Delta_{\mathcal{M}}^{\frac{1}{4}} x \Omega \in \mathcal{H}$ is nuclear.

Problem 5.3. *Is modular nuclearity preserved under finite free product?*

The point is that $\mathcal{N}$ needs to be treated as a Banach space, and one has to estimate the operator norm of the elements x in $\mathcal{N} = \star_{\kappa \in K} (\mathcal{N}_{\kappa}, \omega_{\kappa})$. This is not straightforward, as the very definition of free product depends on the states ω_{κ} .

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